LAB EXPERIMENT
ITA0506-COMPUTER VISION

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EXPNO:21

Aim

To sharpen an image using a Laplacian mask implemented with an extension to diagonal neighbors.

Laplacian Mask

The 8-neighbor Laplacian mask is:

1 1 1

1 -8 1

1 1 1

Here, the four diagonal pixels are also included along with the top, bottom, left, and right pixels.

For sharpening:

Sharpened Image = Original Image - Laplacian Image

Algorithm

1. Read the input image.
2. Convert the image to grayscale.
3. Define the 3×3 Laplacian mask with diagonal neighbors.
4. Apply the mask using convolution.
5. Subtract the Laplacian result from the original image.
6. Clip the pixel values to the range 0–255.
7. Display the original, Laplacian, and sharpened images.

Code:

EXP-21.py - C:/Users/POORNA SKHT/OneDrive/Tài liệu/computer vision ita0502/lab experim...

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```python
import cv2
import numpy as np

# Step 1: Read the image
image = cv2.imread("BEN 10.png")

# Check whether image is loaded
if image is None:
    print("Error: Image not found!")
    print("Make sure BEN 10.png is in the same folder as this Python file.")
    exit()

# Step 2: Convert image to grayscale
gray = cv2.cvtColor(image, cv2.COLOR_BGR2GRAY)

# Step 3: Define Laplacian mask
# Includes diagonal neighbors
mask = np.array([
    [1,  1, 1],
    [1, -8, 1],
    [1,  1, 1]
], dtype=np.float32)

# Step 4: Apply Laplacian mask
laplacian = cv2.filter2D(gray, cv2.CV_32F, mask)

# Step 5: Sharpen the image
sharpened = gray.astype(np.float32) - laplacian

# Step 6: Keep pixel values between 0 and 255
sharpened = np.clip(sharpened, 0, 255)

# Convert to unsigned 8-bit image
sharpened = sharpened.astype(np.uint8)

# Step 7: Convert Laplacian result for display
laplacian_display = cv2.convertScaleAbs(laplacian)

# Step 8: Display images
cv2.imshow("Original Image", gray)
```

Ln: 10 Col: 27

OUTPUT :

Result

The image is successfully sharpened using a Laplacian mask with diagonal neighbors. The inclusion of diagonal neighbors helps detect edges in all eight directions, making fine details and boundaries more prominent.